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1 Abstract

This project presents a comprehensive analysis of Total Harmonic Distortion (THD) in Sinusoidal Pulse Width Modulated (SPWM) inverters, which are widely used for efficient DC to AC power conversion. The study focuses on understanding how variations in modulation index and switching frequency affect the harmonic content in the inverter's output voltage. An SPWM-based inverter system is modeled in MATLAB/Simulink to convert a 350V DC input into a 220V RMS, 50Hz AC output using an H-bridge topology. Simulations are carried out for PWM switching frequencies of 5kHz, 7.5kHz, 10kHz, 15kHz, and 20kHz, across a modulation index range from 0.1 to 1.5. The results include detailed evaluations of THD and normalized fundamental voltage output for each configuration. The analysis identifies the optimal combinations of switching frequency and modulation index that minimize THD while maintaining output voltage quality, with considerations based on the IEEE-519 standard for harmonic distortion. The impact of over-modulation is also examined, highlighting its potential benefits and trade-offs. This study provides valuable insights for optimizing SPWM inverter design to enhance power quality and operational efficiency.

2 Background

2.1 Inverters and Pulse Width Modulation (PWM)

Inverters are electronic devices that convert direct current (DC) into alternating current (AC), enabling the use of DC power sources—such as solar panels or batteries—for AC loads. They are widely utilized in renewable energy systems, uninterruptible power supplies (UPS), electric vehicles, and industrial motor drives. A common topology is the H-bridge, which consists of four switching devices (typically MOSFETs or IGBTs) arranged to control current direction, enabling the generation of both positive and negative halves of an AC waveform.

To synthesize a waveform that resembles a pure sine wave, Pulse Width Modulation (PWM) techniques are employed. PWM involves varying the width of digital pulses at a constant frequency. This modulation allows precise control of the inverter's output voltage and frequency, resulting in improved waveform quality and higher efficiency.[3]

2.2 Sinusoidal Pulse Width Modulation (SPWM)

Sinusoidal PWM (SPWM) is a commonly used method to produce a high-quality AC output in inverters. In SPWM, a sinusoidal reference signal—representing the desired output—is continuously compared with a high-frequency triangular carrier waveform. The instants at which the sinusoidal signal intersects the carrier determine the switching events of the inverter.

This results in a pulse train with variable widths, effectively shaping the output to mimic a sinusoidal waveform. SPWM is especially suitable for applications that require high power quality, such as medical instruments, communication systems, and industrial automation, due to its ability to minimize distortion and improve waveform fidelity

2.3 Total Harmonic Distortion (THD)

Total Harmonic Distortion (THD) is a crucial metric used to assess the waveform quality of inverter output. It quantifies the ratio of harmonic components (frequencies that are integer multiples of the fundamental) to the fundamental frequency itself. Lower THD values indicate cleaner sinusoidal output, which is essential for efficient operation and equipment protection.[1]

In SPWM inverters, THD is influenced by parameters such as the modulation index and switching frequency. High THD can lead to power losses, equipment overheating, and electromagnetic interference. Thus, understanding and

minimizing THD is vital for optimizing inverter performance and ensuring system reliability.[4]

2.4 IEEE-519 Standard and Harmonic Compliance

The IEEE-519 standard provides guidelines for acceptable levels of voltage and current harmonics in electrical power systems. It sets limits on THD based on voltage levels and system configuration to safeguard both utility infrastructure and end-user equipment.

SPWM inverters must be evaluated against these standards to ensure they do not introduce excessive harmonic distortion into the grid. This study adheres to IEEE-519 benchmarks to assess whether different combinations of switching frequencies (5kHz to 20kHz) and modulation indices (0.1 to 1.5) remain within acceptable harmonic limits. Compliance not only ensures system integrity but also supports long-term operational efficiency and regulatory approval.[2]

3 Objective

The primary objectives of this project are:

- Simulate an SPWM inverter in Simulink or PSIM to analyze output voltage characteristics.
- Study the effect of modulation index and PWM frequency on Total Harmonic Distortion (THD).
- Plot THD and normalized fundamental output against varying modulation in dices.
- Identify the optimal modulation index and PWM frequency for minimum THD.
- Understand and apply IEEE-519 standards in analyzing inverter output quality.

4 Problem Statement

The simulation was carried out using MATLAB Simulink for an SPWM inverter with:

- Input Voltage: 350V DC

- The THD was measured using Fast Fourier Transform (FFT) analysis with a limited band selected based on the dominant harmonics visible within 2 kHz beyond the carrier frequency.

5

6 Results

6.1 Graphs

6.1.1 THD vs. Modulation Index Plot

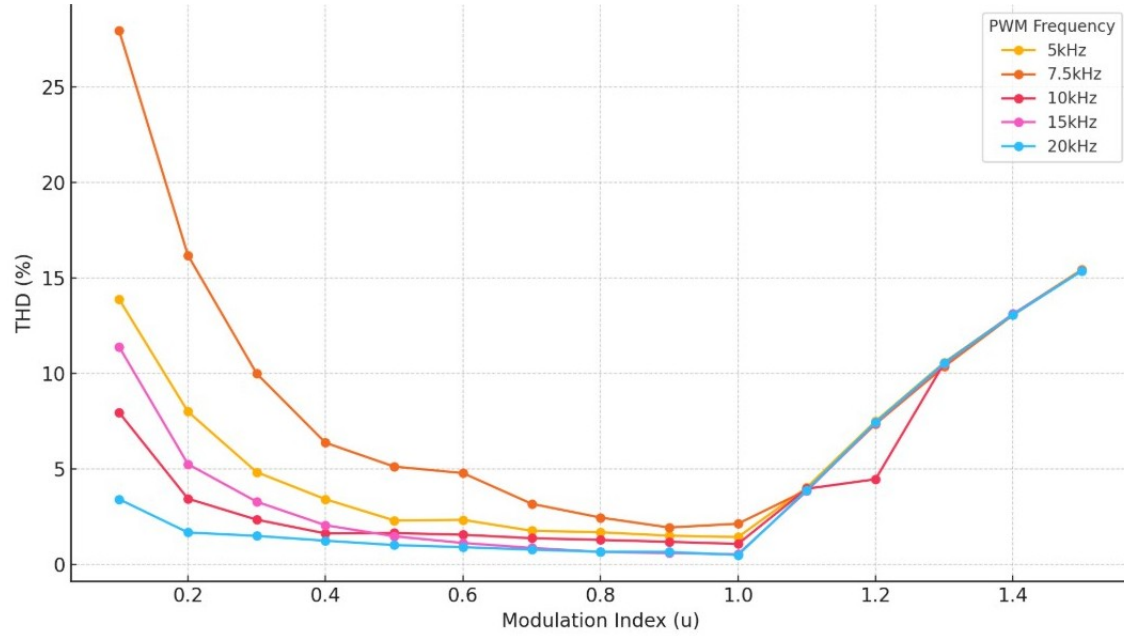


Figure 2: THD vs. Modulation Index Plot for Each PWM Frequency

1. Minimum THD occurs near modulation index = 1.0 for all frequencies.
2. Higher PWM frequencies (15–20 kHz) consistently result in lower THD across all modulation indices.
3. Increases again in the over-modulation region (u greater than 1.0) due to waveform clipping.

6.1.2 THD vs. PWM Frequency Plot

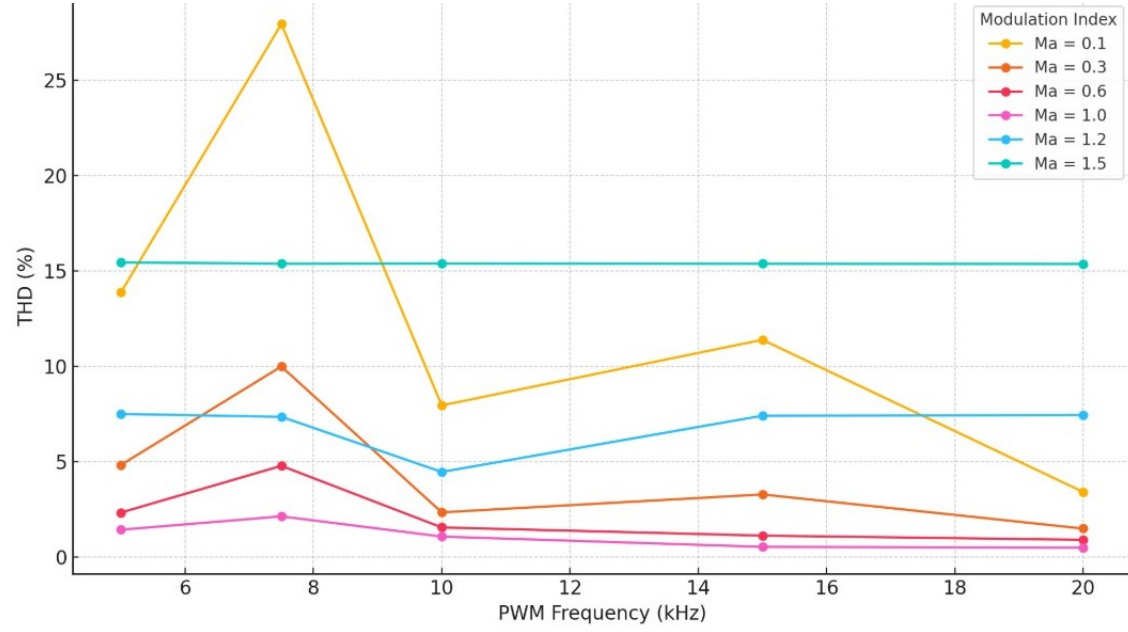


Figure 3: THD vs. PWM Frequency Plot for Selected Modulation Indices

1. Modulation Index = 1.0 shows the lowest THD across all frequencies, especially at 15–20 kHz.
2. $Ma = 0.1$ and $Ma = 1.5$ exhibit high THD, confirming poor harmonic performance in both under- and over-modulation regions.
3. At $Ma = 0.6$, THD steadily decreases with increasing PWM frequency, reflecting the benefit of higher switching frequencies.

6.2 Tables

6.2.1 THD vs. Modulation Index

Table 1: Total Harmonic Distortion (THD %) for Different PWM Frequencies

Modulation Index (μ)	5 kHz	7.5 kHz	10 kHz	15 kHz	20 kHz
0.1	13.89	27.96	7.96	11.40	6.50
0.2	8.01	16.20	3.45	5.26	1.68
0.3	4.84	10.00	2.35	3.29	1.50
0.4	3.42	6.38	1.63	2.06	1.25
0.5	2.31	5.13	1.65	1.48	1.02
0.6	2.34	4.79	1.56	1.13	0.91
0.7	1.77	3.18	1.38	0.87	0.78
0.8	1.69	2.00	1.29	0.66	0.67
0.9	1.51	1.60	1.19	0.60	0.66
1.0	1.44	1.50	1.08	0.54	0.50
1.1	4.04	2.80	3.97	3.85	3.90
1.2	7.51	4.50	4.47	7.42	7.45
1.3	10.59	10.37	10.56	10.51	10.55
1.4	13.12	13.07	13.11	13.13	13.09
1.5	15.46	15.39	15.40	15.39	15.38

6.2.2 THD vs. PWM Frequency

Table 2: THD vs. PWM Frequency for Selected Modulation Indices

PWM Frequency (kHz)	Ma = 0.1	Ma = 0.3	Ma = 0.6	Ma = 1.0	Ma = 1.2	Ma = 1.5
5	13.89	4.84	2.34	1.44	7.51	15.46
7.5	27.96	10.00	4.79	2.14	7.36	15.39
10	7.96	2.35	1.56	1.08	4.47	15.40
15	11.40	3.29	1.13	0.54	7.42	15.39
20	3.42	1.50	0.91	0.50	7.45	15.38

7 Modulation Index M_a Effect on Output Voltage and THD

In sinusoidal PWM inverters, the modulation index (M_a) significantly affects the output voltage and the total harmonic distortion (THD) of the output waveform. The modulation index is defined as:

$$M_a = \frac{V_{\text{ref}}}{V_{\text{carrier}}}$$

7.1 Modulation Index $M_a = 0.1$

Output Voltage:

- Very low output voltage.
- PWM pulses are very narrow.
- The fundamental component is weak.

THD:

- Very high THD.
- Significant harmonic distortion due to narrow pulses.

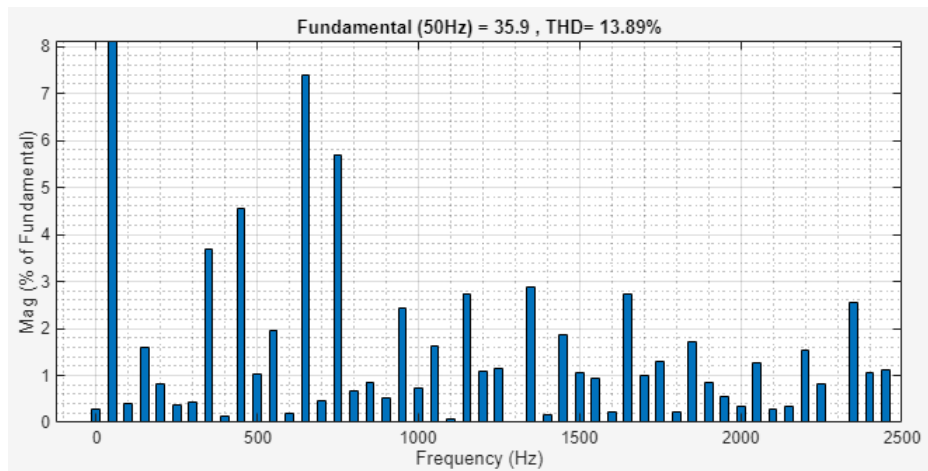


Figure 4: THD at Frequency = 5kHz

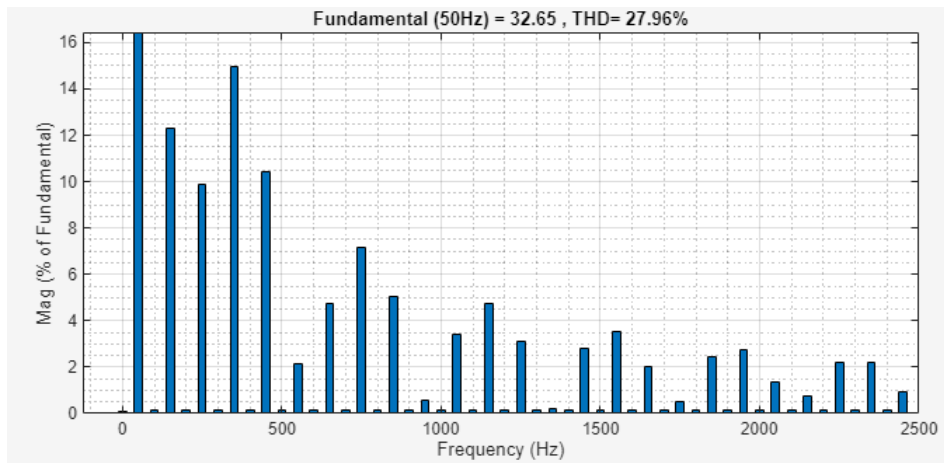


Figure 5: THD at Frequency = 7.5kHz

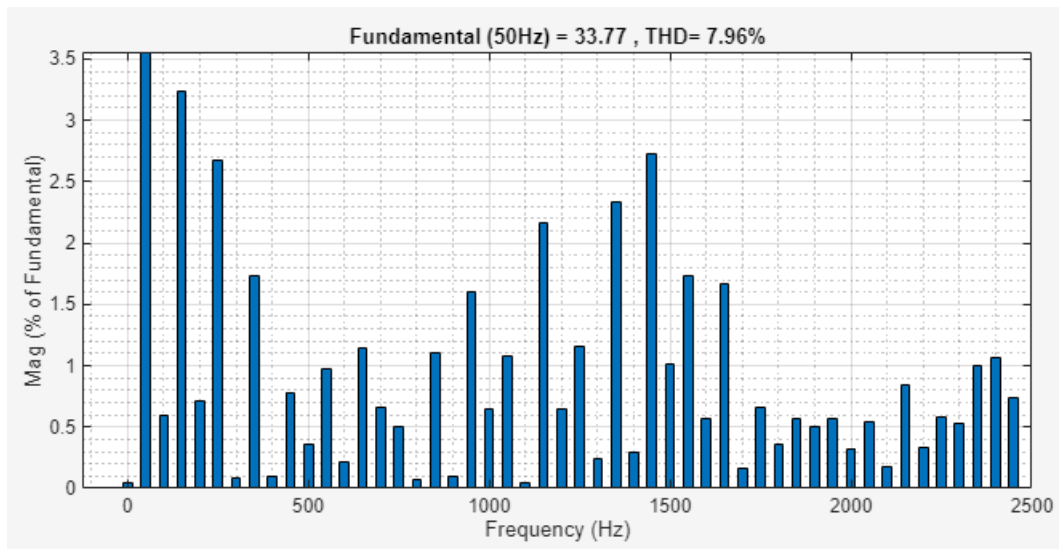


Figure 6: THD at Frequency = 10kHz

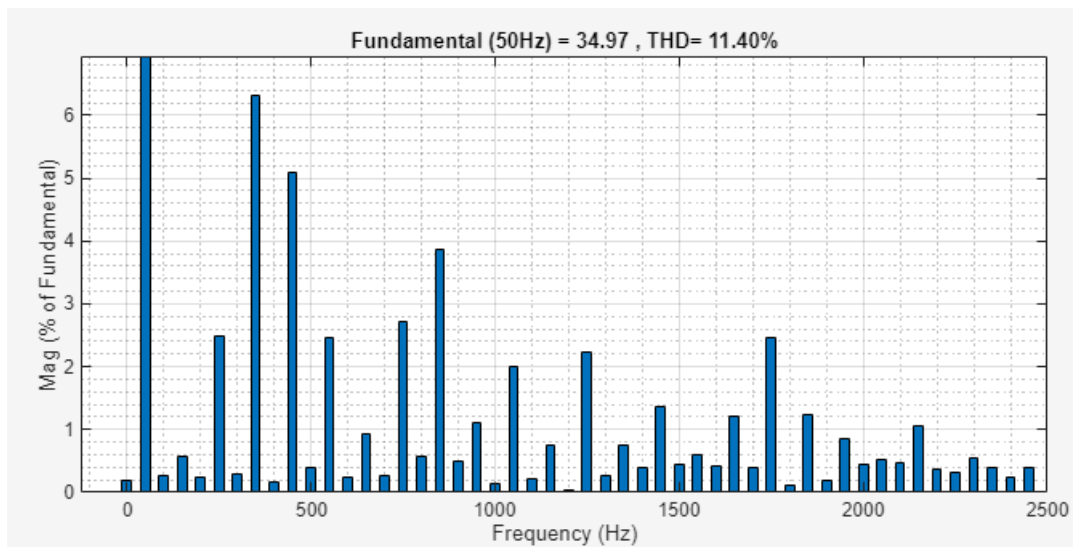


Figure 7: THD at Frequency = 15kHz

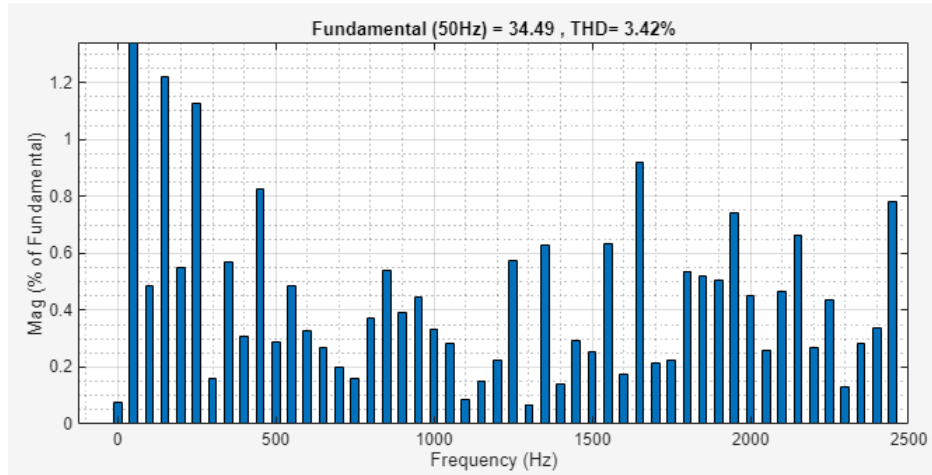


Figure 8: THD at Frequency = 20khz

7.2 Modulation Index $M_a = 0.5$

Output Voltage:

- Moderate output voltage.
- PWM waveform more closely follows the sine wave.

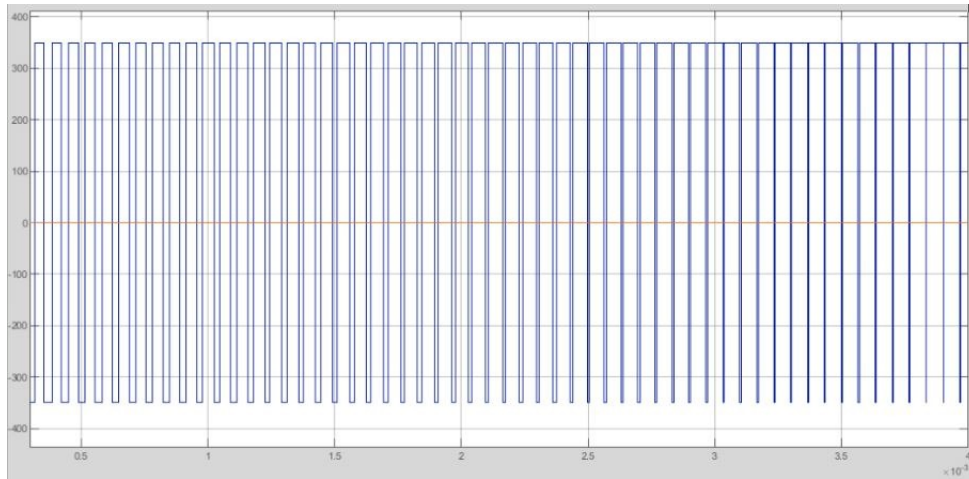


Figure 9: Output Voltage for Modulation Index 0.5

THD:

- Moderate THD.

- Improved waveform quality compared to lower M_a .

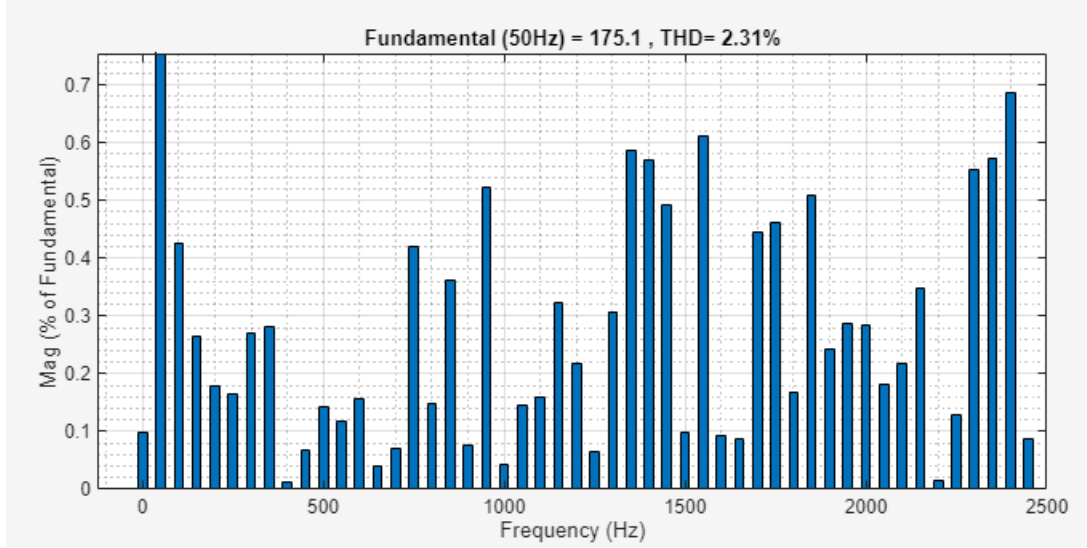


Figure 10: THD at Frequency = 5kHz

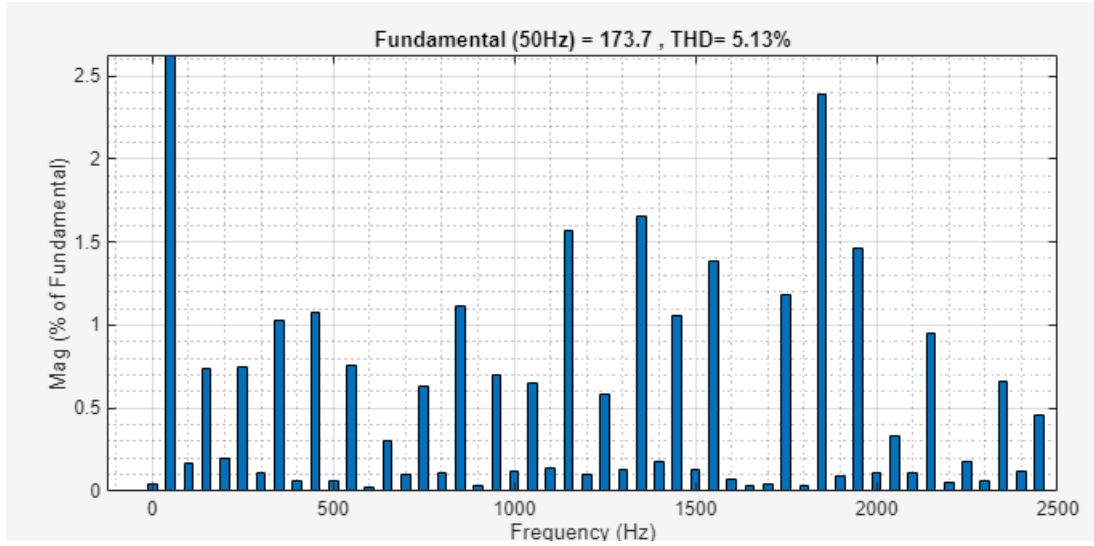


Figure 11: THD at Frequency = 7.5kHz

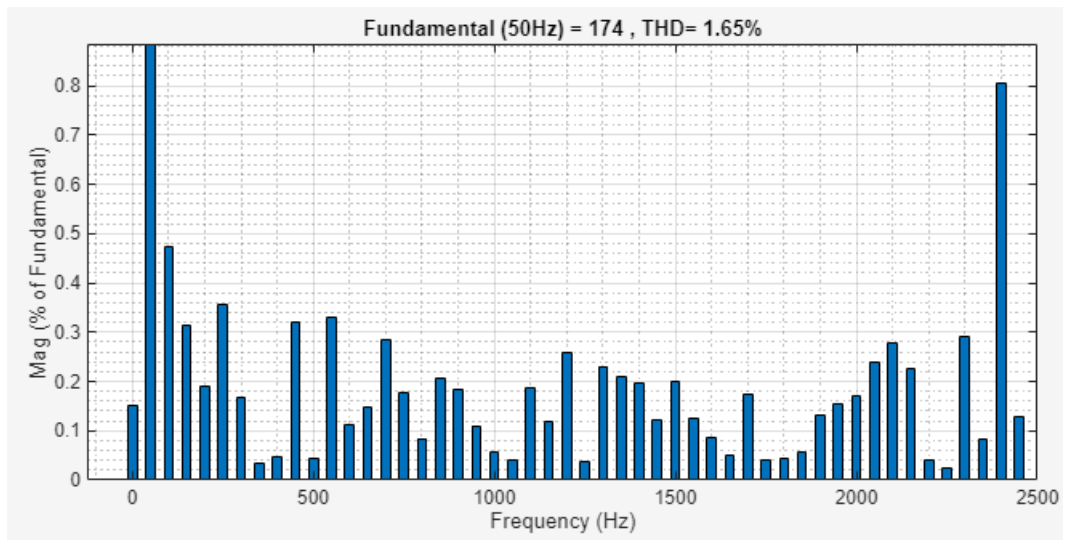


Figure 12: THD at Frequency = 10khz

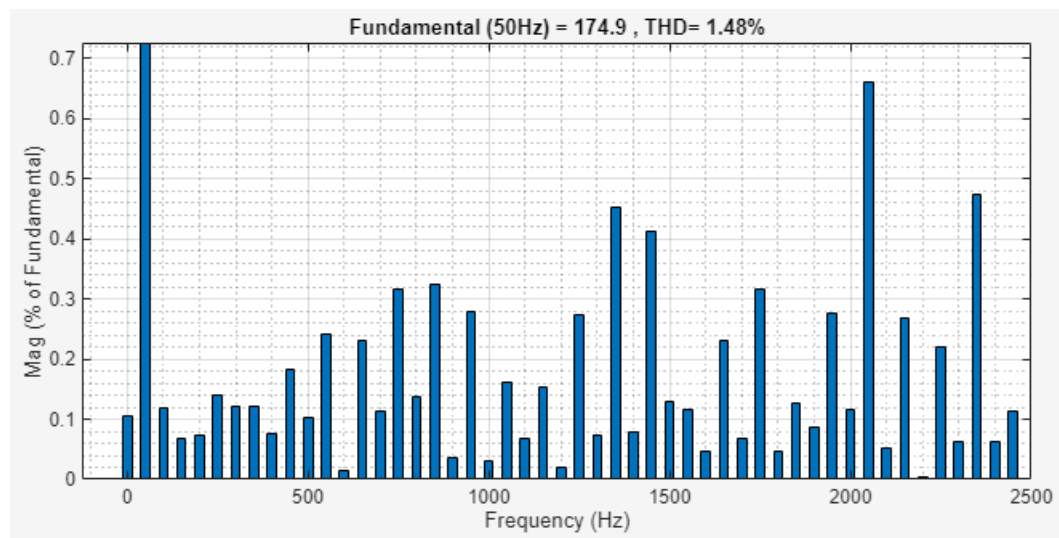


Figure 13: THD at Frequency = 15khz

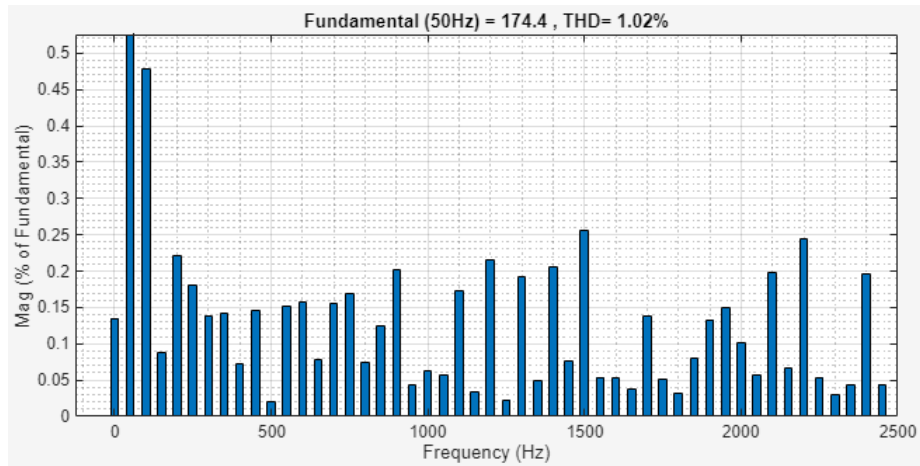


Figure 14: THD at Frequency = 20khz

7.3 Modulation Index $M_a = 1.0$ (Ideal)

Output Voltage:

- Maximum fundamental voltage in the linear range.
- PWM pulses match the sine wave accurately.

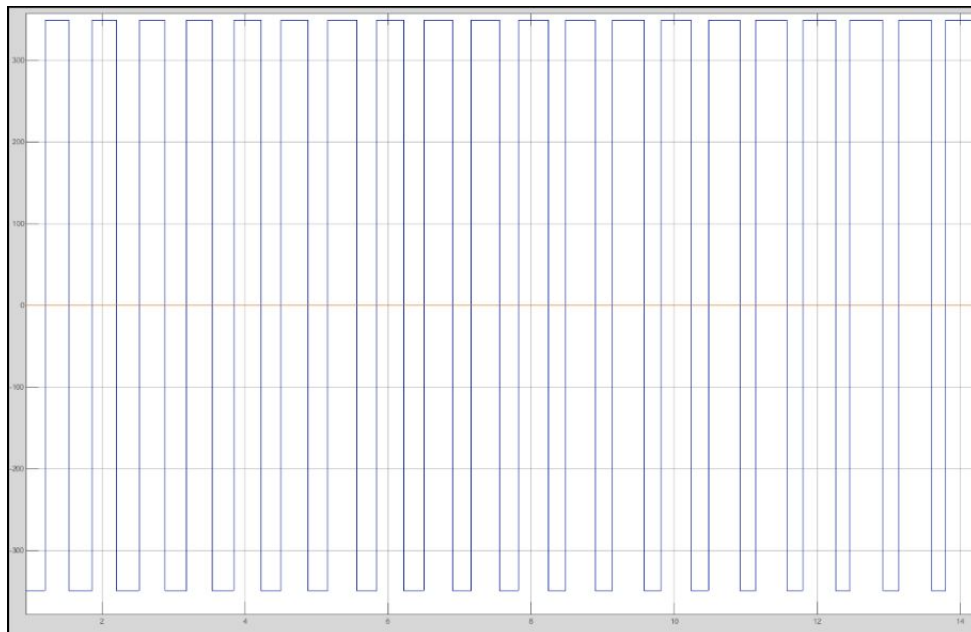


Figure 15: Output Voltage for Modulation Index 1.0

THD:

- Lowest THD.
- Best waveform quality; closest to a pure sine wave.

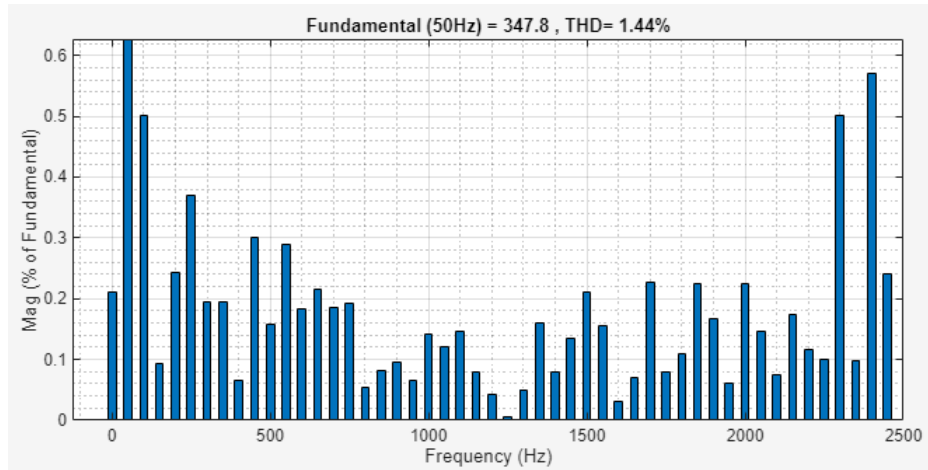


Figure 16: THD at Frequency = 5khz

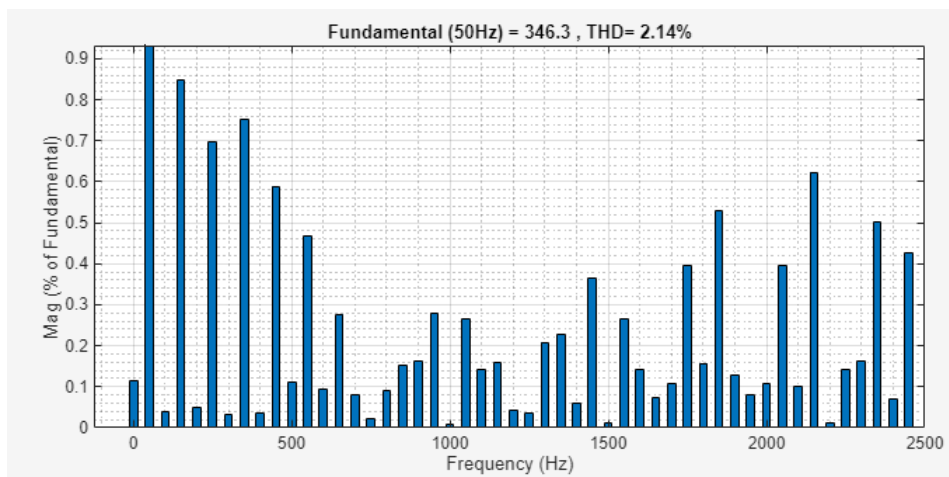


Figure 17: THD at Frequency = 7.5khz

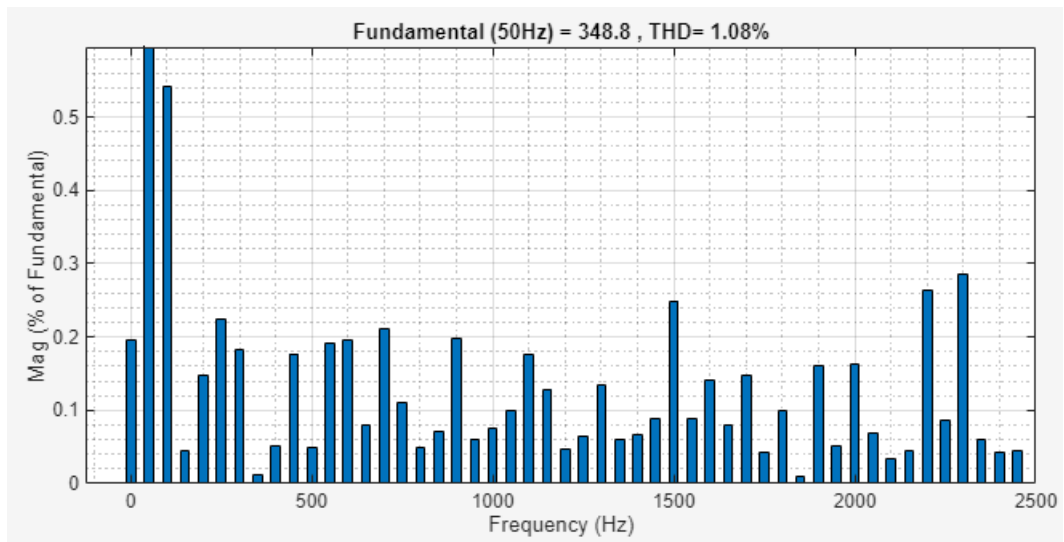


Figure 18: THD at Frequency = 10khz

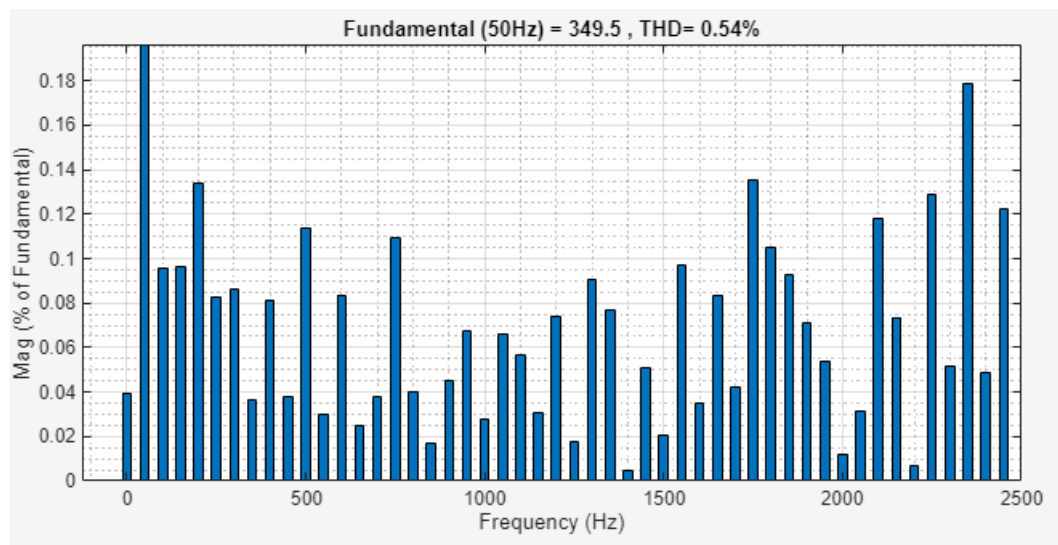


Figure 19: THD at Frequency = 15khz

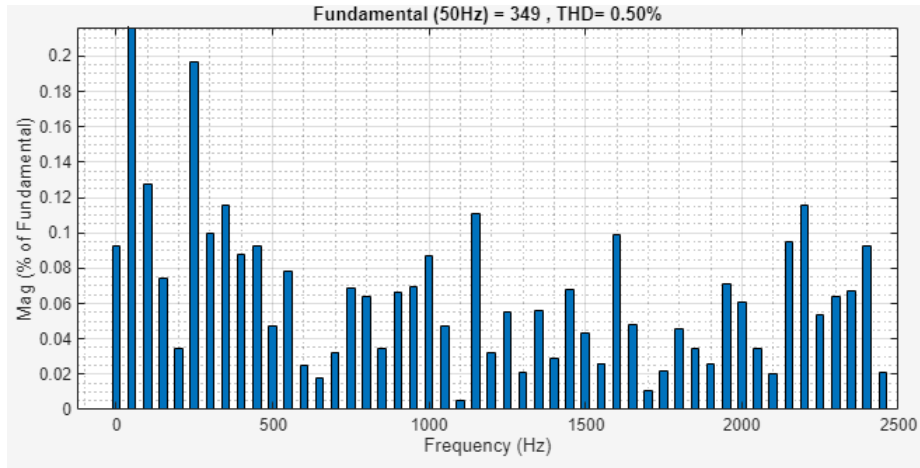


Figure 20: THD at Frequency = 20kHz

7.4 Modulation Index $M_a = 1.5$ (Deep Overmodulation)

Output Voltage:

- Very high output voltage, approaching square wave characteristics.
- Large portion of the waveform is clipped.

THD:

- High THD.
- Output waveform resembles a square wave, with many harmonics.

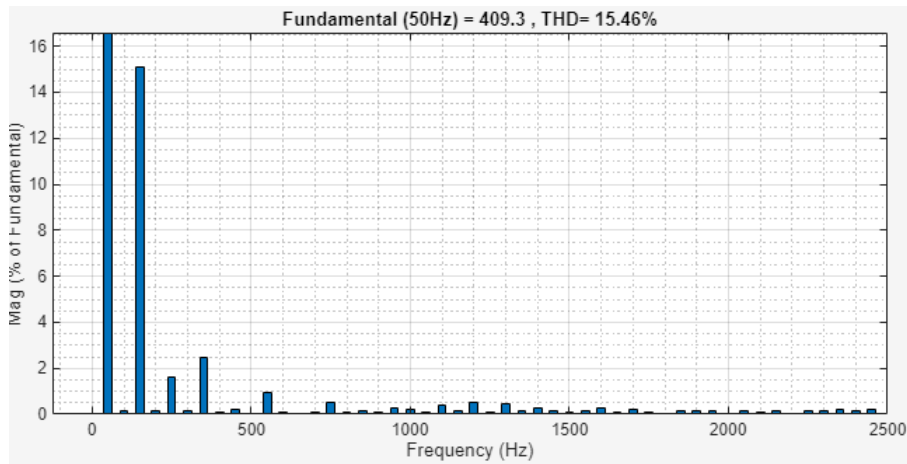


Figure 21: THD at Frequency = 5kHz

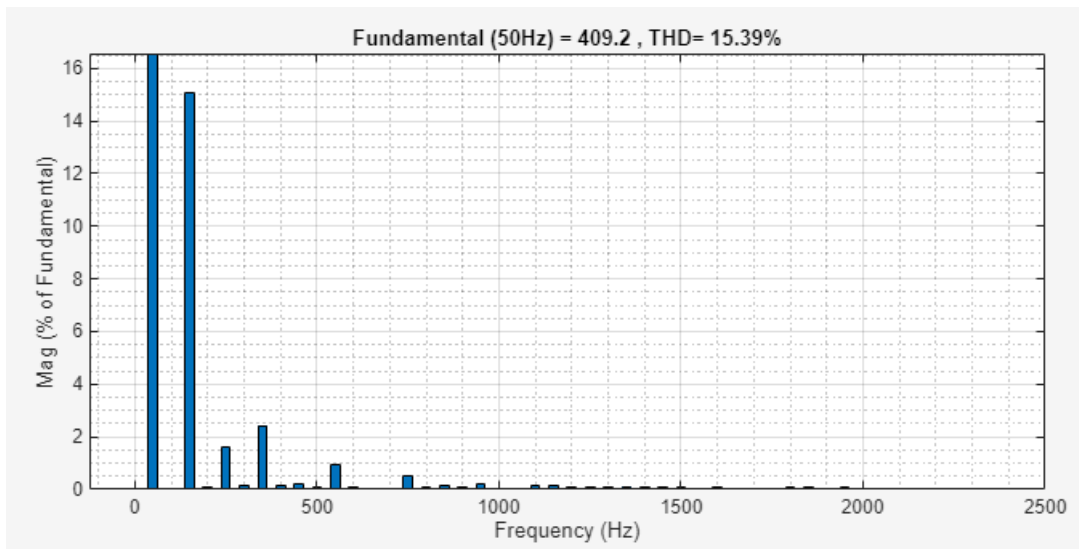


Figure 22: THD at Frequency = 7.5khz

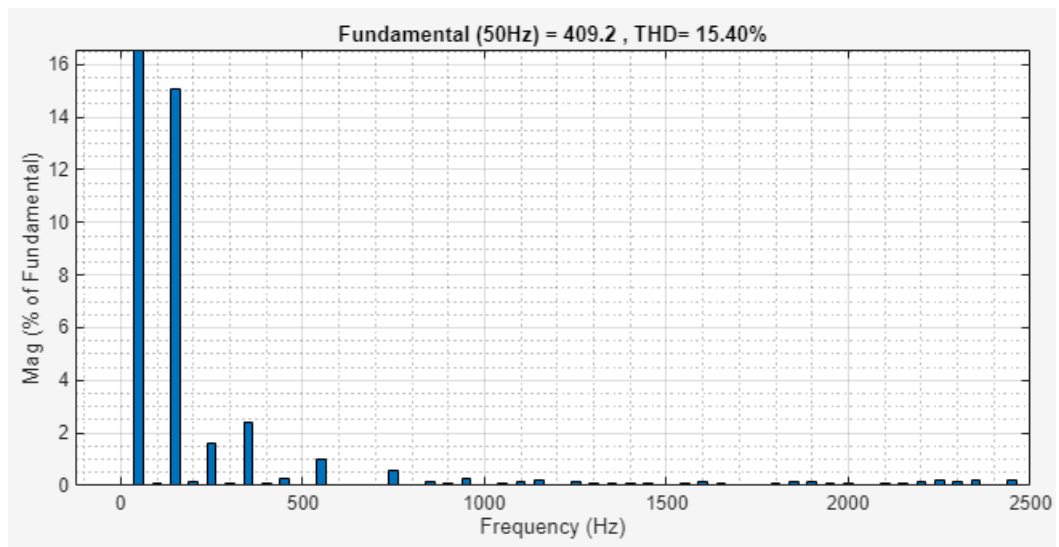


Figure 23: THD at Frequency = 10khz

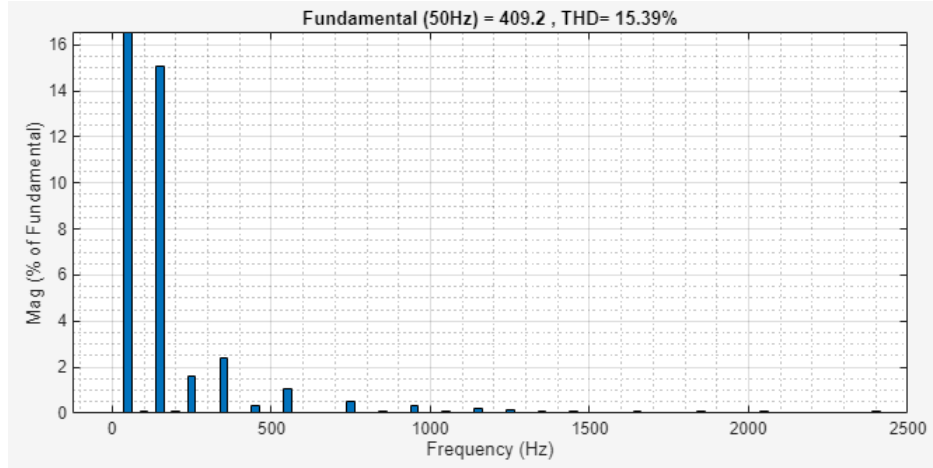


Figure 24: THD at Frequency = 15khz

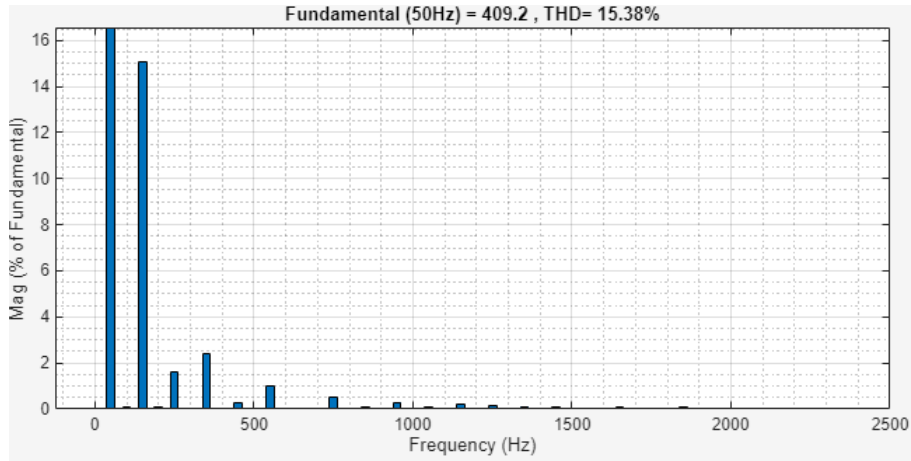


Figure 25: THD at Frequency = 20khz

8 Discussion

The results of this study offer critical insights into how modulation index and PWM frequency influence the Total Harmonic Distortion (THD) and output voltage quality in SPWM inverters. By evaluating both graphical and tabulated data, along with detailed waveform analyses at various modulation indices, it becomes evident that proper tuning of these parameters is essential to optimize inverter performance and comply with IEEE-519 harmonic standards.

The graphical analysis presented in Figures 2 and 3 shows a clear correlation between THD and both modulation index and PWM frequency. In Figure 2, THD decreases with increasing modulation index, reaching a minimum around $Ma = 1.0$, beyond which THD increases again due to waveform clipping in the over-modulation region. Figure 3 reinforces the advantage of using higher PWM frequencies—specifically 15kHz to 20kHz—as they consistently produce lower THD across most modulation indices. These plots effectively demonstrate that both under- and over-modulation contribute significantly to harmonic distortion, while optimal performance lies in the linear modulation region.

The tabulated results (Tables 1 and 2) provide a numerical representation of THD trends. Table 1 clearly shows how increasing PWM frequency leads to lower THD across all modulation indices, especially at $Ma = 1.0$, where THD drops to as low as 0.50 at 20kHz. Table 2 tells this, showing minimal THD at higher frequencies and ideal modulation indices. These values validate the graphical observations and help identify optimal inverter configurations that balance output quality and harmonic performance.

At $Ma = 0.1$, the inverter operates in the deep under-modulation region, producing very narrow PWM pulses and a weak fundamental output voltage. As seen in Figures 4 to 8, the resulting waveforms are heavily distorted, and THD is extremely high across all PWM frequencies. This configuration is inefficient for practical applications due to poor voltage quality and excessive harmonic content.

With $Ma = 0.5$, output voltage increases and the PWM waveform begins to resemble a sine wave more closely. Figures 9 to 14 show improved waveform symmetry and moderately reduced THD. While not optimal, this setting offers a reasonable compromise between output voltage and harmonic content, making it more suitable than $Ma = 0.1$ but still not ideal for sensitive applications.

$Ma = 1.0$ represents the ideal operating point of the SPWM inverter. As shown in Figures 15 to 20, the output voltage waveform aligns almost perfectly with a sine wave, and THD is minimized across all frequencies, particularly at 20kHz where it drops to just 0.50. This configuration provides the best spectral performance and meets IEEE-519 harmonic standards, making it the most desirable choice for high-quality power applications.

Operating at $Ma = 1.5$ leads to overmodulation, where the output waveform begins to resemble a square wave due to significant clipping. This is evident in Figures 21 to 25, which show increased THD at all frequencies. While the output voltage is maximized, the harmonic performance deteriorates substantially. This trade-off may be acceptable in applications that demand higher voltage but are tolerant to harmonic distortion.

9 Comments

9.1 THD vs Modulation Index

- THD decreases with increasing modulation index in the linear range ($Ma \leq 1.0$).
- Minimum THD is consistently observed at $Ma = 1.0$ across all PWM frequencies.
- THD rises sharply in the over-modulation region ($Ma > 1.0$) due to waveform clipping.

9.2 Impact of PWM Frequency

- Higher PWM frequencies result in lower THD values for all modulation indices.
- Best waveform quality observed at 15kHz and 20kHz due to finer resolution in PWM control.
- Although not simulated here, high switching frequencies can lead to increased switching losses.

9.3 Best Operating Point

- The best THD performance occurs at $Ma = 1.0$ and PWM frequencies of 15kHz–20kHz.
- This point yields THD values below 1%, meeting IEEE-519 standards.
- Optimal for applications requiring high power quality and minimal distortion.

9.4 Output Voltage Spectrum

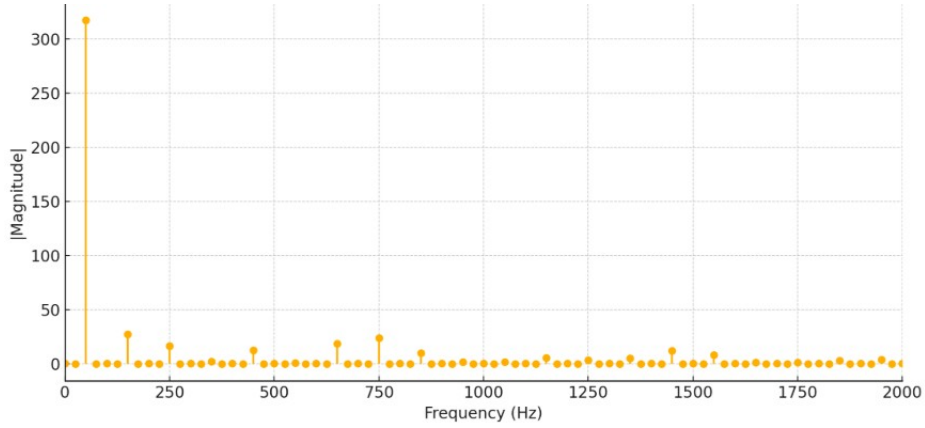


Figure 26: Output Voltage Spectrum at Modulation Index = 1.0 and PWM Frequency = 20kHz

- At $Ma = 1.0$ and 20kHz PWM, the output shows a strong 50Hz fundamental component.
- Harmonic components above the carrier frequency are significantly attenuated.
- High-frequency PWM helps shift harmonics to a range where filtering is easier.

9.5 Over-Modulation Region

- In over-modulation ($Ma > 1.0$), THD increases sharply due to significant waveform clipping.
- Output begins to resemble a square wave, increasing the presence of higher-order harmonics.
- Suitable only for applications where higher voltage is needed and harmonic distortion is acceptable.

10 Conclusion

The simulation and analysis confirm that both modulation index and PWM switching frequency play vital roles in shaping the output waveform and determining the

Total Harmonic Distortion (THD) of SPWM inverters. An ideal modulation index of $Ma = 1.0$ combined with a high switching frequency (15kHz or 20kHz) consistently produces the lowest THD, achieving values well below 1% and aligning with IEEE-519 harmonic limits. These conditions also result in a waveform that closely matches a pure sine wave, making them suitable for sensitive AC applications. Conversely, operation in under-modulation (e.g., $Ma = 0.1$) or over-modulation regions (e.g., $Ma = 1.5$) significantly increases THD, compromising output quality. While over-modulation provides higher output voltage, the trade-off is excessive harmonic distortion. Therefore, inverter design should prioritize operating within the linear modulation region and selecting appropriate PWM frequencies to balance efficiency, voltage regulation, and harmonic compliance. This study provides a solid foundation for optimizing SPWM inverters for both academic and industrial power electronics applications.

References

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